

Physical Safety

Stealthy multi-modal attack surfaces in Mixed Reality:

AIXVR'24, SenSys'20 (poster),

Under preparation: exploring additional surfaces

Tools & Techniques: sensors, in-the-wild data collection

HoloSet: A dataset for Visual -inertial Pose Estimation:

DATA'22 (ACM SenSys)

Under consideration: adding hand and eye data

Tools & Techniques: in-the-wild data collection, Unity 3D

Robustness Analysis of SLAM Tracking:

Under review: ACM TSAS

Tools & Techniques: image processing, empirical analysis

Neruosymbolic Adaptive Feature Extraction:

Under preparation: ECCV'24

Tools & Techniques: domain-specific abstraction,
mathematical modeling, fitness function

Dataset Characterization as an Input Profiler:

Under preparation: gathering validation samples

Tools & Techniques: image processing, time-series
data analytics

Cognitive Safety

Reaction Time as a Quantitative Measure of Presence:

IEEE TVCG

Tools & Techniques: Unity 3D, user study, psychology

Human Factor Influencing Presence Measures:

IEEE VR'24

Tools & Techniques: Unity 3D, user study,
psychology

Interaction Affecting Presence and Reaction Time:

Under preparation: ISMAR'24

Tools & Techniques: Unity 3D, user study, psychology

Privacy Leakage in Collaborative Mixed Reality :

Under preparation: building a theoretical foundation

Tools & Techniques: sensor data, Unity 3D, user study,
psychology

**Individual User Behavior in Collaborative Mixed
Reality:**

Under preparation: ISMAR'24

Tools & Techniques: sensor data, Unity 3D, user study,
psychology, time-series data analytics

